

Table 1. Summary of tested working fluids and surfaces within this study.

Sample Name	Working Fluid	Concentration (wt.%)	Static Contact Angle (°)	
			Before Boiling	After Boiling
REF	Water	/	61	93
SHPI	Water	/	0	110
SHPO	Water	/	164	148
SHPI-0.001%	TiO ₂ -water	0.001	0	17
SHPI-0.1%	TiO ₂ -water	0.1	0	0
SHPO-0.001%	TiO ₂ -water	0.001	164	79
SHPO-0.1%	TiO ₂ -water	0.1	164	20

The examination of functionalized surface properties involved utilizing scanning electron microscopy (SEM) and contact angle measurements. To analyze the micro- and nanostructure of the surfaces, a scanning electron microscope (ThermoFisher Scientific Quattro S, Waltham, MA, USA) was employed, employing both backscattered electron (BSE) and secondary electron (SE) imaging at an accelerating voltage of 10 kV. Figure 1 shows SEM images of the untreated and laser-engineered surfaces before they underwent boiling tests. Contact angle measurements were conducted using a goniometer (Ossila, Sheffield, UK), and the assessments were carried out with twice-distilled water at room temperature. These measurements were performed on the prepared and tested samples both immediately before and after the boiling test within the scope of this study. The static contact angle results for the tested surfaces are presented in Table 1.

